

controllers. In addition to its standard products, ST offers customers the possibilities of application-specific devices and access to leading-edge silicon processes and package technologies for customer-designed products to give added performance and value to the end product.

Opportunities galore

In an emerging market like India, there is a lot of potential for growth and demand for medical electronics (of which semiconductors are an essential part). The increasing awareness of healthcare has created a rapidly growing need for innovative medical solutions like portable and affordable ultrasound equipment for use in doctors' equipment, ambulances, mobile triage solutions and remote regions. Responding to this demand, Indian electronics OEMs are developing a number of portable devices like digital thermometers, blood pressure monitors, insulin pumps, heart rate monitors and digital hearing aids in addition to hospital equipment like CT and MRI scanners, X-ray machines and ultrasound scanners.

Semiconductors are set to play a pivotal role in designing and developing portable and affordable medical devices. The connectivity enabled by semiconductor technology (wired or wireless) will drive such applications as telemedicine, facilitating access to the required level of healthcare.

Challenges

With the given constraints of large population residing in the villages and doctors living in urban areas, the challenge is to make the healthcare delivery accessible and affordable. The technology advancements and innovations that are happening in this space are helping to overcome this challenge. Devices are becoming affordable and portable aided by the fact that chips are getting packed with more functionalities. Mobile technology is helping overcome the accessibility challenge.

Still, there are challenges to overcome, like lack of electronic medical

Major ongoing projects under Medical Electronics & Telemedicine Division, Department of Information Technology, Ministry of Communications & Information Technology, government of India

1. Deployment of indigenously developed 6MeV medical LINAC for cancer treatment—Jai Vigyan Phase II—by SAMEER, Mumbai
2. Design and development of electronic portal imaging device (EPID) for radiation therapy by CSIO, Chandigarh, and TSG, Integration, New Delhi
3. Establishment of facility for batch fabrication of LINAC tube and machines by SAMEER, Mumbai
4. Design and development of 2.6MW S-Band tunable pulse magnetron by CEERI, Pilani (Rajasthan)
5. Design and development of dual-photon energy and multiple-electron energy integrated oncology system by SAMEER, Mumbai
6. Design and development of multileaf collimator for use in medical linear accelerator machine by SAMEER, Mumbai
7. Development of high-speed inter-point braille embosser by WML, Kolkata, and CMERI, Durgapur
8. Design and development of cost-effective bio-signals-controlled prosthetic hand by Tezpur University
9. Technology development for building distributed, scalable and reliable healthcare information store by CDAC, Pune
10. Development of medical image analyser for cervical cancer (Cervi SCAN) by CDAC, Thiruvananthapuram, IIT, Kharagpur and RCC, Thiruvananthapuram
11. Virtual-reality-based minimally-invasive surgical simulator (VR-MISS) with haptics feedback by IIT, Chennai, and CMC, Vellore
12. Development of a Web-enabled e-healthcare system for neonatal patient care services (eNPCS) by IIT, Kharagpur, SSKM Hospital, Kolkata, and WECS, Kolkata
13. Web-enabled medical information access using handheld devices in a wireless environment for telemedicine applications by IIT, Kharagpur, and WECS, Kolkata
14. Development of telemedicine at remote CHC/PHC in Tripura by WECS, Kolkata
15. Development and implementation of ICT-based mobile tele-oncology system for extending the coverage of ONCONET, Kerala, by CDAC, Thiruvananthapuram
16. National resource centre for telemedicine and medical informatics by SGPGL, Lucknow

records and matured ecosystem in India for more device innovations. With growing popularity of medical electronics, there is now a growing concern on quality, safety, reliability, liability and regulatory issues.

Indian medical device manufacturers have to compete with foreign players who have an open entry in India. Contrary to that, entry is not as easy for Indian companies who wish to promote their products outside India.

The complete Indian medical devices market, more so for patient monitoring systems, is mainly driven by costs. The industry experiences a spurt in demand with a drop in prices of medical devices. In fact, Chinese players have played a key role in the sphere of cost reduction, though at the

cost of quality.

Time is right for an Indian standard for quality to be put in place for any device to be used in India and products not conforming to these standards should not be allowed for local consumption.

Future outlook

The future seems to be in how the devices integrate themselves in a hospital ecosystem and their ability to communicate with other systems. Equipment having the ability to transmit data through telemedicine are now available, though telemedicine is confined to more of visual communication devices as of now. ●

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